# House Price Prediction Report

This report summarizes the predictions generated by the machine learning model trained on the housing dataset. The system predicts house prices using features such as bedrooms, bathrooms, square footage, and year built.

## Dataset Summary

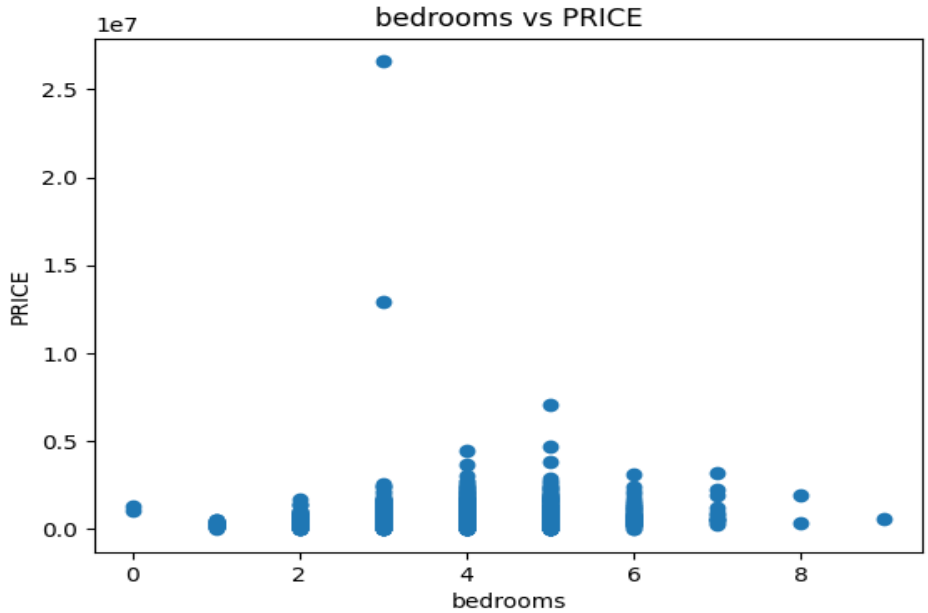
| Dataset Detail | Value |
|---|---|
| Total Records | 4600 |
| Total Features | 17 |
| Target Variable | price |

## Features Used for Prediction

- bedrooms
- bathrooms
- sqft_living
- sqft_lot
- floors
- waterfront
- view
- condition
- sqft_above
- sqft_basement
- yr_built
- yr_renovated

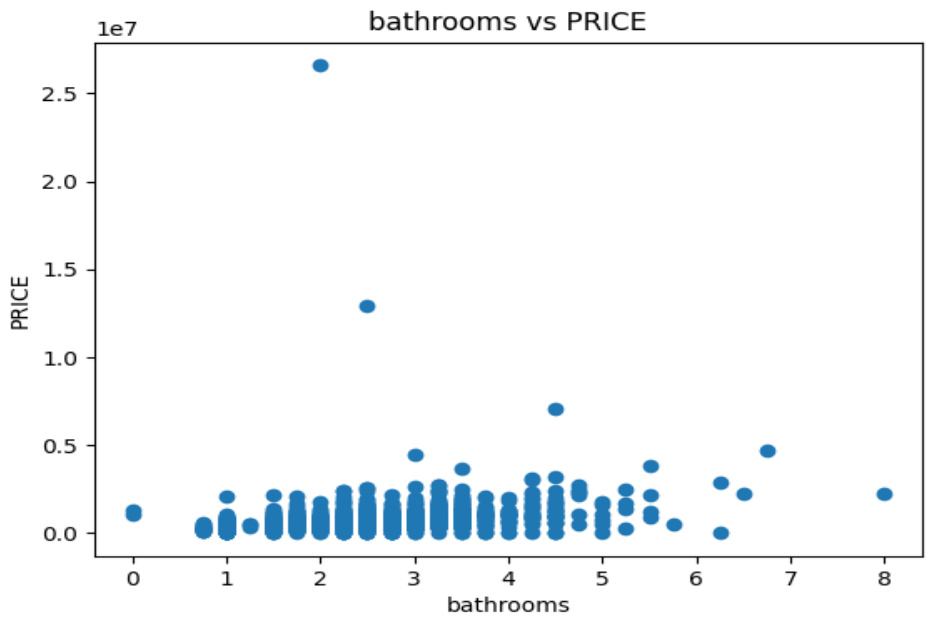
## Feature vs Price Scatter Plots

bedrooms vs PRICE



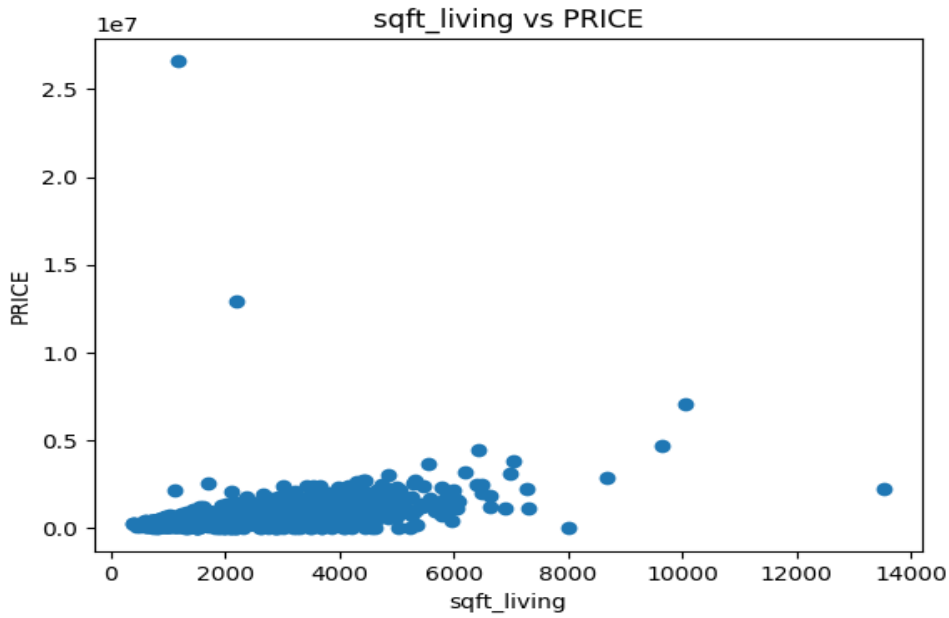

This scatter plot shows the relationship between the feature **bedrooms** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **bedrooms** may influence the predicted price of houses.

bathrooms vs PRICE



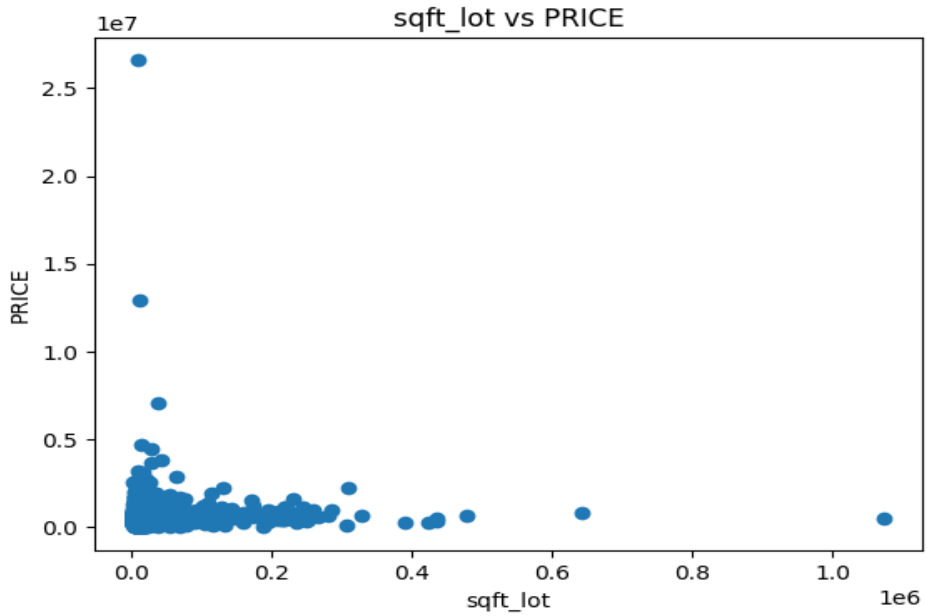

This scatter plot shows the relationship between the feature **bathrooms** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **bathrooms** may influence the predicted price of houses.

sqft_living vs PRICE




This scatter plot shows the relationship between the feature **sqft_living** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **sqft_living** may influence the predicted price of houses.

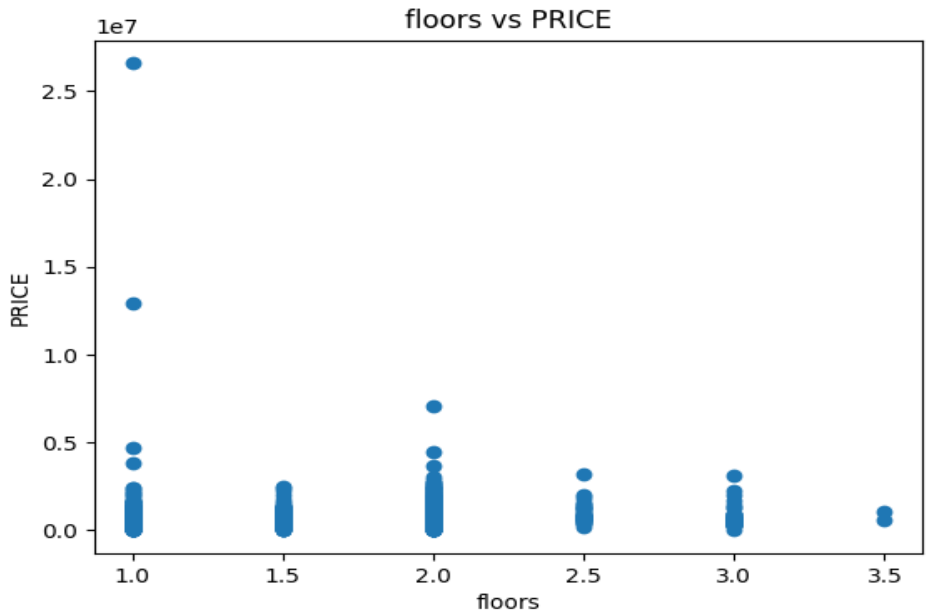
sqft_lot vs PRICE




This scatter plot shows the relationship between the feature **sqft_lot** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a

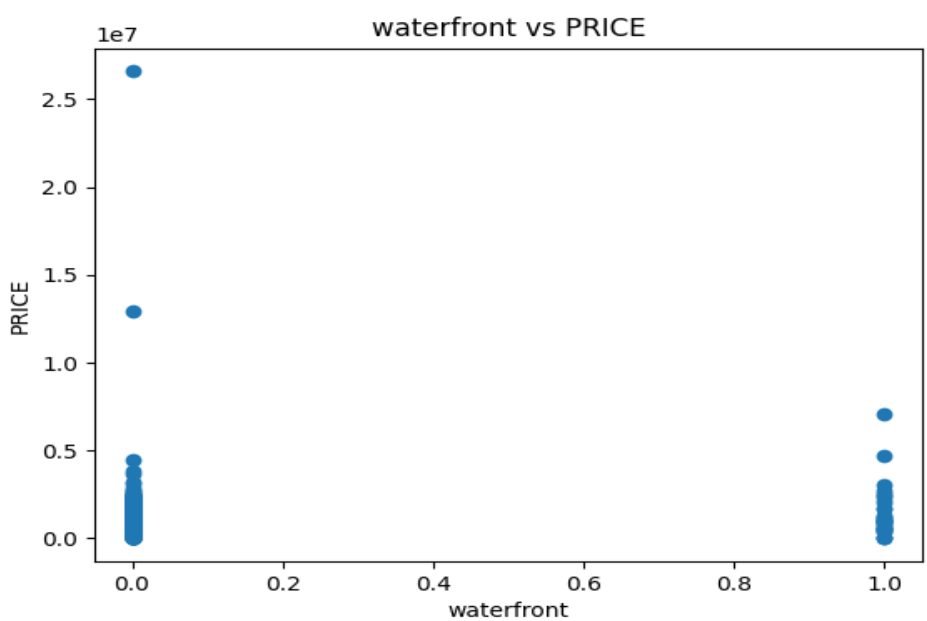
relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **sqft_lot** may influence the predicted price of houses.

floors vs PRICE



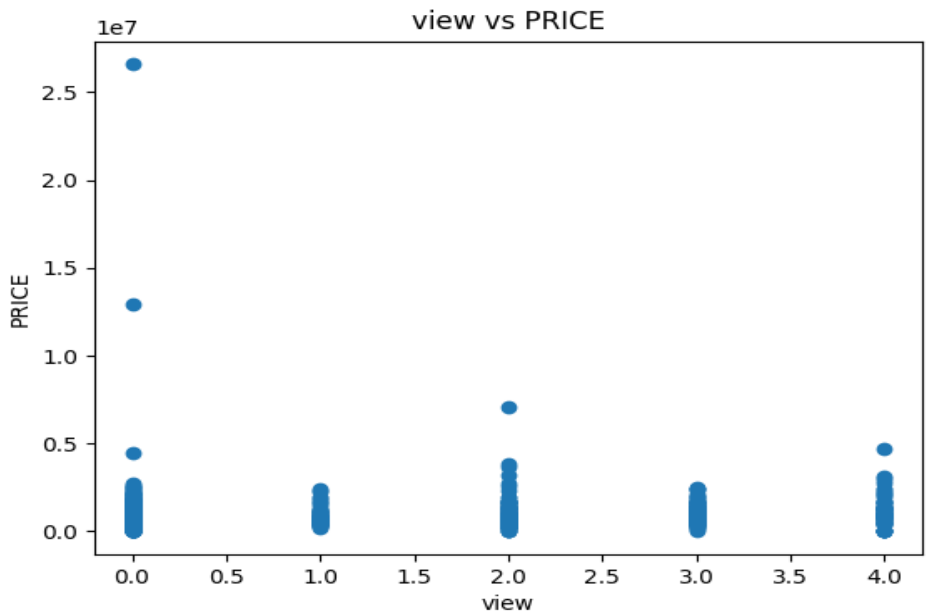

This scatter plot shows the relationship between the feature **floors** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **floors** may influence the predicted price of houses.

waterfront vs PRICE



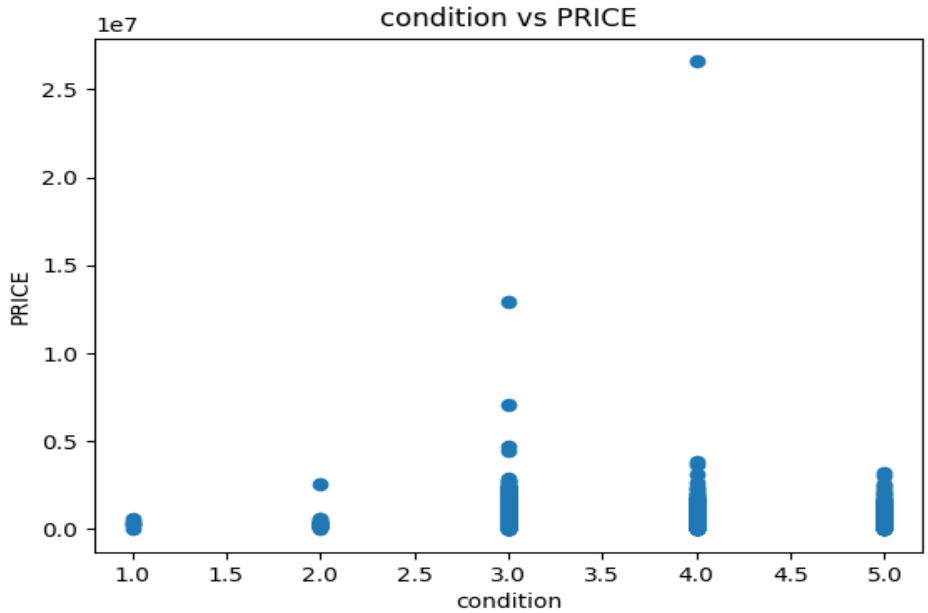
This scatter plot shows the relationship between the feature **waterfront** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **waterfront** may influence the predicted price of houses.

view vs PRICE



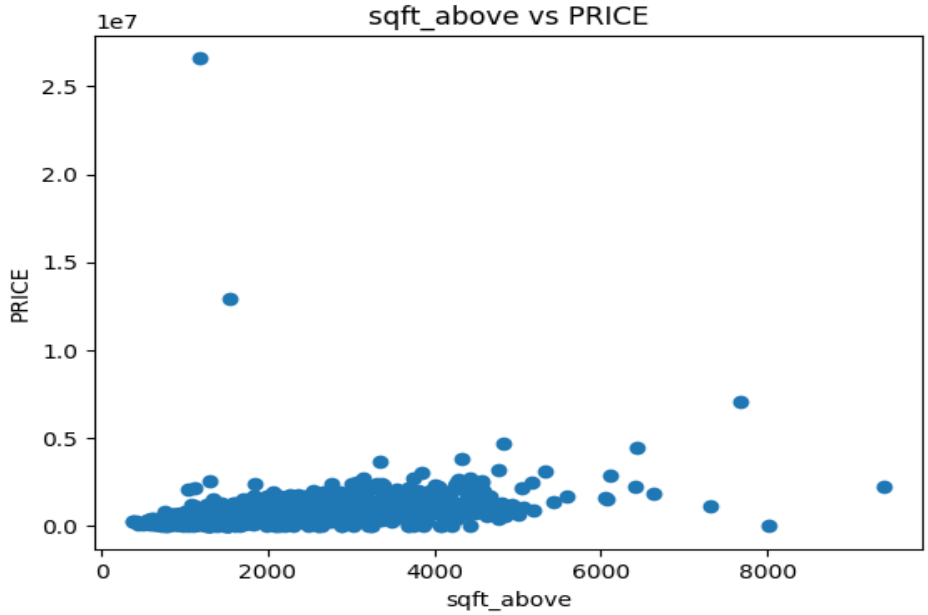

This scatter plot shows the relationship between the feature **view** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **view** may influence the predicted price of houses.

condition vs PRICE



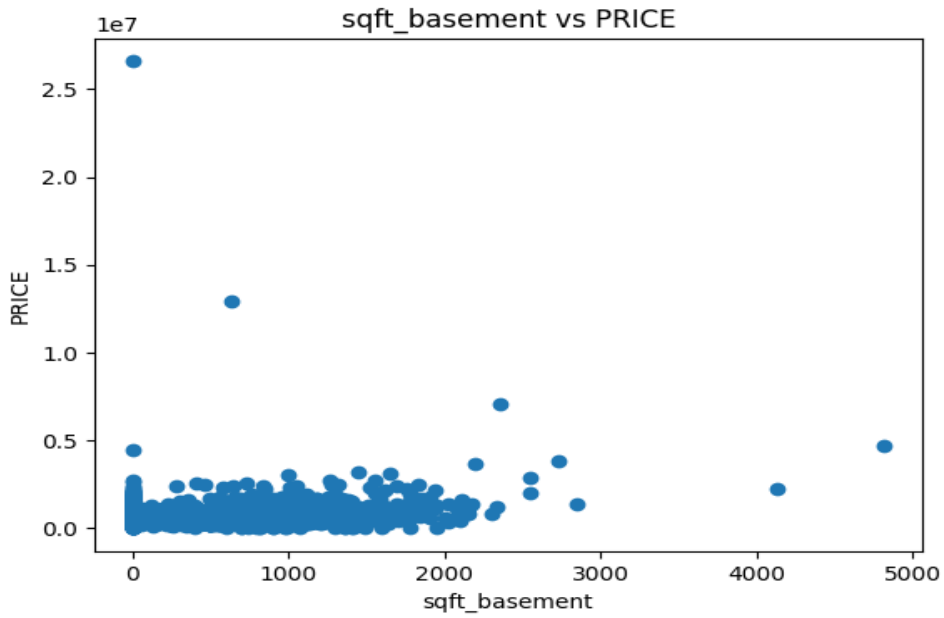

This scatter plot shows the relationship between the feature **condition** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **condition** may influence the predicted price of houses.

sqft_above vs PRICE



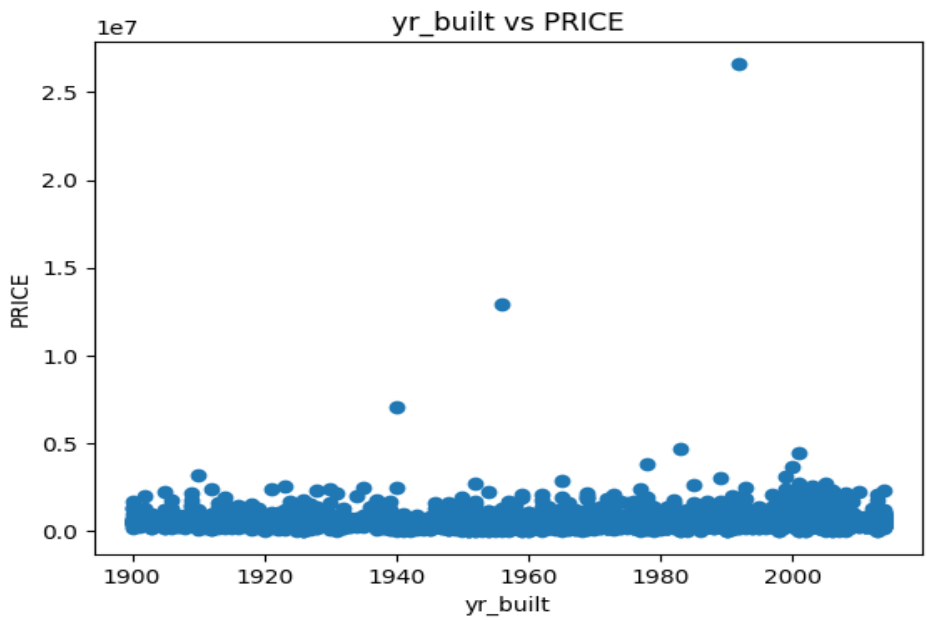

This scatter plot shows the relationship between the feature **sqft_above** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **sqft_above** may influence the predicted price of houses.

sqft_basement vs PRICE




This scatter plot shows the relationship between the feature **sqft_basement** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **sqft_basement** may influence the predicted price of houses.

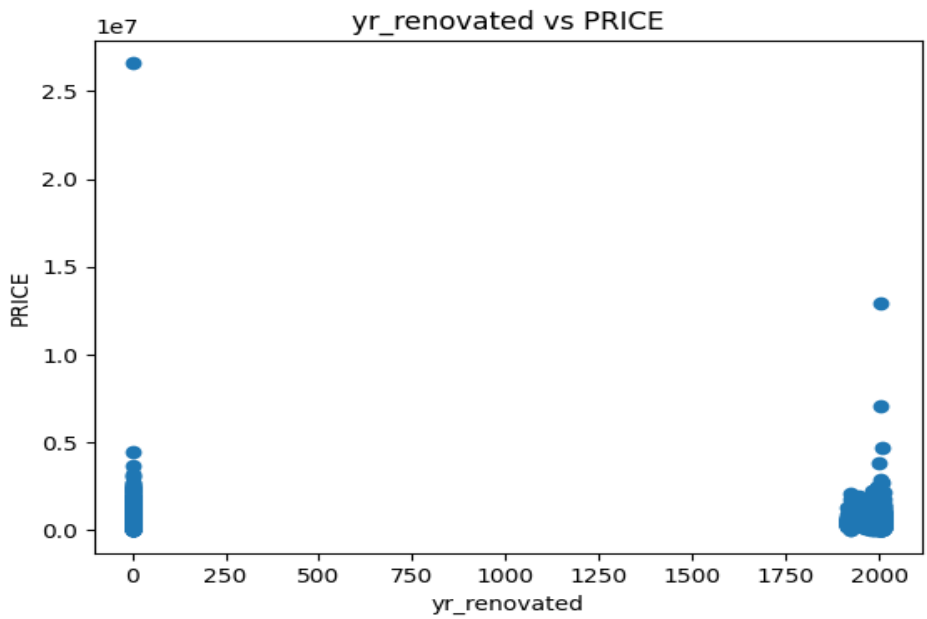
yr_built vs PRICE




This scatter plot shows the relationship between the feature **yr_built** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a

relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **yr_built** may influence the predicted price of houses.

yr_renovated vs PRICE




This scatter plot shows the relationship between the feature **yr_renovated** and the target variable **PRICE**. Each point represents an individual house from the dataset. Scatter plots help visualize whether a relationship exists between the feature and house prices. Observing the distribution of points allows us to understand how variations in **yr_renovated** may influence the predicted price of houses.

## Model Information

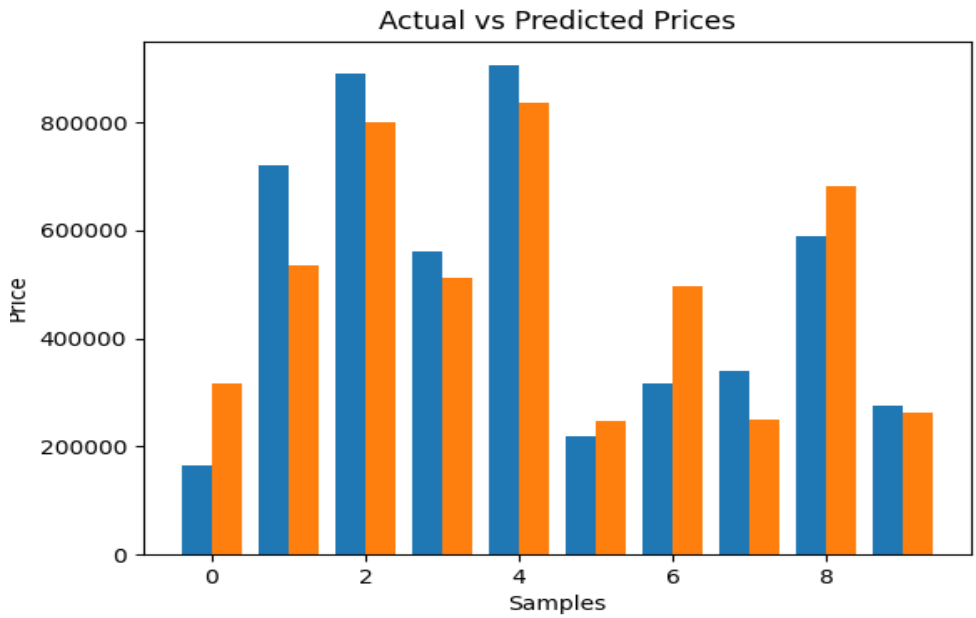
Best Model Selected: Linear Regression
R-Squared Score: -0.4804

## Actual vs Predicted Prices

| Actual Price | Predicted Price |
|---|---|
| 163785.71 | 316232.4 |
| 720000.0 | 534914.92 |
| 890000.0 | 800491.36 |
| 562100.0 | 513390.08 |
| 905000.0 | 834933.9 |
| 218000.0 | 248530.24 |
| 318000.0 | 496108.5 |

| | |
|---|---|
| 340000.0 | 250371.65 |
| 589900.0 | 681367.0 |
| 276000.0 | 261615.0 |

## Prediction Visualization




This bar chart compares the actual house prices from the dataset with the prices predicted by the machine learning model. The visualization helps evaluate how closely the model's predictions match the real values. If the predicted bars are close to the actual bars, it indicates that the model is performing well in estimating house prices.

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